

# Lumio

## Clinical Intelligence for Mental Health

### Google Calendar & Dual-Calendar Sync Architecture Specification

#### 1. EXECUTIVE SUMMARY & KEY CAPABILITIES

- **Dual-Calendar Synchronization:** Saves to Custom App DB (data/events.json) first, then pushes to User's Google Calendar.
- **Instant Google Meet Provisioning:** Generates unique https://meet.google.com URLs automatically.
- **Multi-Guest Direct Invitations:** Sends official Google Calendar email invites to multiple clients directly from the app, bypassing the standard "Add Guest" flow.
- **Seamless Google OAuth 2.0 SSC:** Captures offline refresh tokens for background calendar interaction.

#### DATA FLOW ARCHITECTURE DIAGRAM

The Lumio Google Calendar & Dual-Calendar Sync Engine provides automated, HIPAA-compliant event scheduling, multi-guest invitation handling, and instant video link generation. It synchronizes meeting data simultaneously across two distinct destinations:

- 1. Custom Application Database (data/events.json)
- 2. User's Google Calendar (via OAuth 2.0 SSC)

#### 2. SYSTEM ARCHITECTURE & SEQUENCE FLOW

- [ Stage 1: Commit Local DB ] (data/events.json Custom View)
- [ Stage 2: Call Google Calendar API ] (Generates Meet Link & Emails Guests)

#### 3. TECHNOLOGY STACK & DEPENDENCIES

| Layer             |                               |                                                      |
|-------------------|-------------------------------|------------------------------------------------------|
| Technology        |                               | Purpose & Architectural Role                         |
| Framework         | Next.js 15 (App Router)       | SSR, API routes, Server Actions & Turbopack          |
| Authentication    | NextAuth.js (Auth.js v5)      | Google OAuth 2.0 with offline access tokens          |
| Google SDK        | googleapis v144               | Direct integration with Google Calendar API v3       |
| Database          | JSON File Store (events.json) | In-app custom calendar event persistence             |
| UI Styling        | Tailwind CSS v4               | Custom Lumio warm-cream palette & responsive layouts |
| Icons & Utilities | Lucide React, UUID, Zod       | Vector icons, unique request IDs & validation        |

- **API Enabled:** Google Calendar API v3
- **OAuth Scopes:** openid, profile, email, <https://www.googleapis.com/auth/calendar.events>
- **Authorized Origins:** <http://localhost:3001>, <https://google-calander-hv6k04gxy-vanco.vercel.app>

#### 4. GOOGLE CLOUD PLATFORM (GCP) CONFIGURATION

**Redirect URIs:** <http://localhost:3001/api/auth/callback/google>  
**// app/actions/create-event.ts - 3 Stage Meeting Creation & Google Calendar Sync** <https://google-calander-hv6k04gxy-vanco.vercel.app/api/auth/callback/google>

- **Publishing Mode:** External (Test users added: kishorbrindha18@gmail.com)
- To enable authentication and calendar synchronization, the GCP Project must be configured with the following parameters:

**LIVE VERCEL PRODUCTION URL:** <https://google-calander-hv6k04gxy-vanco.vercel.app>

**GITHUB REPOSITORY:** <https://github.com/kishorsumathi/lumio-calendar-poc>

#### 5. DEPLOYMENT CREDENTIALS & LIVE ENDPOINTS

**ENVIRONMENT VARIABLES (VERCEL & LOCAL):** AUTH\_SECRET, AUTH\_GOOGLE\_ID, AUTH\_GOOGLE\_SECRET configured.

#### 6. KEY CODE IMPLEMENTATION MODULES

#### 7. TROUBLESHOOTING & ERROR RESOLUTIONS

| Error / Symptom                  |                                                   |                                                                                                                         |
|----------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Root Cause                       |                                                   | Resolution                                                                                                              |
| Error 400: redirect_uri_mismatch | Callback URL in app does not match GCP Console.   | Add <a href="http://localhost:3001/api/auth/callback/google">http://localhost:3001/api/auth/callback/google</a> to GCP. |
| Error 403: access_denied         | GCP App is in Testing mode & email is not listed. | Add testing email under OAuth Consent Screen > Test Users.                                                              |
| Broken Google Avatar Picture     | Google CDN blocks requests with referrer headers. | Use <code>&lt;img referrerPolicy="no-referrer" /&gt;</code> with fallback badge.                                        |
| No Google Meet Link              | conferenceDataVersion parameter missing.          | Set conferenceDataVersion: 1 in calendar.events.insert call.                                                            |



